

Replay-Aware ATSP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: atsp_no_replay.
- Best TSPLIB ATSP holdout gap: atsp_no_replay.
- Best synthetic holdout gap: atsp_no_replay.

Run Metadata

- run_name: run_20260513_160339_a
- started_at_local: 2026-05-13 16:03:39
- finished_at_local: 2026-05-13 16:15:48
- duration_hhmm: 00:12
- duration_seconds: 729.074
- seed_offset: 0
- replicate_label: a
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB ATSP Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
atsp_no_replay	none	score_only	False	0.190767	0.009004	0.099886	0.0	0.5775	0.0
atsp_random_replay	random	score_only	False	0.207155	0.030781	0.118968	0.0	0.78	0.0
atsp_failure_replay	failure	score_only	False	0.207396	0.014741	0.111069	0.809219	0.78	0.044372
atsp_failure_replay_compression	failure	novelty_gate	True	0.193143	0.009004	0.101074	0.830791	0.58	0.012299

Condition Notes

atsp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.190767, synthetic 0.009004, combined 0.099886.

- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.58.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.190767 across 4 instances; family means: ft=0.322375, ftv=0.33478, p=0.003915, ry=0.101997.
- Panel synthetic_holdout mean gap 0.009004 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.002199.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1939, "family": "ftv", "name": "ftv38", "optimality_gap": 0.26732}, {"best_known_cost": 1286, "cost": 1537, "family": "ftv", "name": "ftv33", "optimality_gap": 0.195179}, {"best_known_cost": 1473, "cost": 1705, "family": "ftv", "name": "ftv35", "optimality_gap": 0.157502}]

atsp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.207155, synthetic 0.030781, combined 0.118968.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.78.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.207155 across 4 instances; family means: ft=0.307748, ftv=0.292622, p=0.002135, ry=0.226113.
- Panel synthetic_holdout mean gap 0.030781 across 4 instances; family means: clockwise_ring=0.085613, corridor_drift=0.0, hub_spokes=0.037511, wind_clusters=0.0.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1980, "family": "ftv", "name": "ftv38", "optimality_gap": 0.294118}, {"best_known_cost": 1286, "cost": 1619, "family": "ftv", "name": "ftv33", "optimality_gap": 0.258942}, {"best_known_cost": 1473, "cost": 1756, "family": "ftv", "name": "ftv35", "optimality_gap": 0.192125}]

atsp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.207396, synthetic 0.014741, combined 0.111069.
- Accepted-epoch count 2, mean accepted code novelty 0.809219, and final complexity 0.78.
- Adaptation efficiency 0.044372 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.207396 across 4 instances; family means: ft=0.321796, ftv=0.358958, p=0.001068, ry=0.14776.
- Panel synthetic_holdout mean gap 0.014741 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.040637, wind_clusters=0.018328.
- Worst recent training cases: [{"best_known_cost": 1473, "cost": 1981, "family": "ftv", "name": "ftv35", "optimality_gap": 0.344874}, {"best_known_cost": 1530, "cost": 2039, "family": "ftv", "name": "ftv38", "optimality_gap": 0.33268}, {"best_known_cost": 1286, "cost": 1629, "family": "ftv", "name": "ftv33", "optimality_gap": 0.266719}]

atsp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB ATSP 0.193143, synthetic 0.009004, combined 0.101074.
- Accepted-epoch count 4, mean accepted code novelty 0.830791, and final complexity 0.58.
- Adaptation efficiency 0.012299 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.

- Panel heldout_tsplib mean gap 0.193143 across 4 instances; family means: ft=0.323244, ftv=0.33478, p=0.004093, ry=0.110456.
- Panel synthetic_holdout mean gap 0.009004 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.002199.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1903, "family": "ftv", "name": "ftv38", "optimality_gap": 0.243791}, {"best_known_cost": 1473, "cost": 1749, "family": "ftv", "name": "ftv35", "optimality_gap": 0.187373}, {"best_known_cost": 1286, "cost": 1478, "family": "ftv", "name": "ftv33", "optimality_gap": 0.1493}]

Judge Appendix

Replay-aware ATSP Benchmark Suite Analysis

Condition	Replay Mode	Final T SPLIB Gap	Final Sy nthetic Gap	Final Transfer Gap	Code Novelty (mean)	Co mpl exit y	Notes on Transfer & Novelty
atsp_no_replay	none	0.1908	0.0090	0.0999	0.0	0.58	Best gaps overall on TSPLIB, synthetic, and transfer; no replay, zero code novelty.
atsp_random_replay	random	0.2072	0.0308	0.1190	0.0	0.78	Transfer and synthetic gaps worse than no-replay; no code novelty improvement despite higher complexity.
atsp_failure_replay	failure	0.2074	0.0147	0.1111	0.81	0.78	Code novelty increased substantially but transfer (TSPLIB) performance degraded vs. no replay; synthetic gap improved over random replay but not no replay.
atsp_failure_replay_compression	failure + co mpression	0.1931	0.0090	0.1011	0.83	0.58	High code novelty with compression; transfer metrics close to no replay but slight degradation; complexity reduced via compression.

Interpretation

- **Transfer Performance**:
The **no replay** condition shows superior transfer performance as measured by held-out TSPLIB gap (0.1908) and synthetic holdout gap (0.0090), outperforming all replay conditions.
- **Code Novelty vs Transfer**:
Conditions employing **failure replay** (with and without compression) yield substantially higher code novelty (mean ~0.81–0.83), but this does **not** translate to improved transfer metrics, which are slightly worse than no replay.
- **Complexity and Compression Effects**:

Failure replay with compression maintains high code novelty but reduces complexity (0.58 vs. 0.78), showing that compression pressure can restore complexity levels closer to no replay while preserving novelty. This condition nearly matches no replay in transfer gaps but does not surpass it.

- **Random Replay**:

Random replay results in the worst transfer gaps among the conditions, despite zero code novelty and highest complexity (0.78). This suggests random replay may impair transferability.

- **Replay Mode Effects**:

- **No replay** (atasp_no_replay) is best for transfer metrics with zero code novelty.
- **Failure replay** induces novel code but leads to transfer degradation.
- **Random replay** degrades transfer without increasing novelty.
- **Compression-aware failure replay** improves complexity but does not fully recover no replay transfer performance.

Summary

- The **no replay** condition achieves the best transfer and synthetic holdout optimality gaps with no increase in code novelty.
- Replay mechanisms (random or failure) increase complexity and/or code novelty but do not improve transfer; often transfer gaps worsen.
- Compression-aware failure replay mitigates complexity growth and closes the transfer gap somewhat but does not surpass no replay.
- Therefore, **higher code novelty through replay does not equate to improved algorithmic transferability** in this benchmark.
- The evidence favors **conservative interpretation** that replay and associated code novelty provide no clear transfer advantage over no replay.